Seals & O-Rings

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
NBR O-Ring; EPDM O-Ring	IS 9975 (Rubber O-Rings); IS 5382; IS 7466; IS 11149; IS 10188; IS 13137; IS 3400	ISO 3601-1 (Fluid Power Systems — O-rings — Inside Diameters, Cross-sections, Tolerances and Size Identification Code); ISO 1817 (Rubber — Determination of Effect of Liquids),ISO 3601-3; ISO 3601-5; ISO 4633; ISO 6447; ISO 9631	ASTM D1414 (Rubber O-Rings — Test Methods); ASTM D2000 (Rubber Products Classification System); ASTM D471 (Effect of Liquids on Rubber); ASTM D2240 (Durometer Hardness of Rubber); ASTM C1115; ASTM C361; ASTM C443; ASTM C564; ASTM C864; ASTM D1869; ASTM D5193; ASTM F104; ASTM F112; ASTM F1276; ASTM F146; ASTM F152; ASTM F2524; ASTM F36; ASTM F37; ASTM F38; ASTM F433; ASTM F495; ASTM F607; ASTM F806	DIN ISO 3601-1 (Fluid Systems — O-rings — Dimensions and Tolerances); DIN ISO 3601-5 (O-rings — Quality Acceptance Criteria); DIN ISO 815-1 (Compression Set of Rubber); DIN ISO 1817 (Resistance of Rubber to Liquids)	SAE AS568 (Aerospace O-Ring Sizes); SAE J200 (Rubber Material Classification); NSF/ANSI 61 (Water Applications, where applicable)
Silicone O-Ring	IS 9975 (Rubber O-Rings); IS 14967 (Silicone Rubber Applications); IS 5382; IS 7466; IS 11149; IS 10188; IS 13137; IS 3400	ISO 3601-1 (Fluid Power Systems — O-rings — Inside Diameters, Cross-sections, Tolerances and Designation); ISO 3601-2 (Housing Dimensions); ISO 3601-3 (Quality Acceptance Criteria),ISO 3601-5; ISO 4633; ISO 6447; ISO 9631	ASTM D1414 (Rubber O-Rings — Test Methods); ASTM D2000 (Rubber Products Classification System); ASTM D471 (Effect of Liquids on Rubber); ASTM D2240 (Durometer Hardness of Rubber); ASTM C1115; ASTM C361; ASTM C443; ASTM C564; ASTM C864; ASTM D1869; ASTM D5193; ASTM F104; ASTM F112; ASTM F1276; ASTM F146; ASTM F152; ASTM F2524; ASTM F36; ASTM F37; ASTM F38; ASTM F433; ASTM F495; ASTM F607; ASTM F806	DIN ISO 3601-1 (O-rings — Dimensions and Tolerances); DIN ISO 815-1 (Compression Set); DIN ISO 188 (Accelerated Ageing Resistance); DIN ISO 37 (Tensile Properties of Rubber)	FDA 21 CFR 177 (Food Contact Silicone); USP Class VI (Medical Applications); SAE AS568 (O-Ring Dimensions)
FKM (Viton) O-Ring	IS 9975 (Rubber O-Rings); IS 5382; IS 7466; IS 11149; IS 10188; IS 13137; IS 3400	ISO 3601-1 (Fluid Power Systems — O-rings — Dimensions and Tolerances); ISO 1817 (Rubber — Determination of Effect of Liquids),ISO 3601-3; ISO 3601-5; ISO 4633; ISO 6447; ISO 9631	ASTM D1414 (Rubber O-Rings — Test Methods); ASTM D2000 (Rubber Products Classification System); ASTM D471 (Effect of Liquids on Rubber); ASTM D1418 (Rubber and Rubber Latices — Nomenclature); ASTM C1115; ASTM C361; ASTM C443; ASTM C564; ASTM C864; ASTM D1869; ASTM D5193; ASTM F104; ASTM F112; ASTM F1276; ASTM F146; ASTM F152; ASTM F2524; ASTM F36; ASTM F37; ASTM F38; ASTM F433; ASTM F495; ASTM F607; ASTM F806	DIN ISO 3601-1 (O-rings — Dimensions); DIN ISO 1817 (Resistance to Fluids); DIN ISO 815-1 (Compression Set); DIN ISO 188 (Heat Ageing Resistance)	SAE AS568 (O-Ring Sizes); SAE J200 (Rubber Classification); NORSOK M-710 (Oil & Gas Elastomer Qualification)
Hydraulic Seal	IS 15466:2004; IS 5382; IS 7466; IS 11149; IS 10188; IS 13137; IS 3400	ISO 7425-1 (Hydraulic Fluid Power — Housing Dimensions for Elastomeric Seals); ISO 5597 (Pneumatic Fluid Power — Rod Seals, where applicable),ISO 3601-1; ISO 3601-3; ISO 3601-5; ISO 4633; ISO 6447; ISO 9631	ASTM D2000 (Rubber Products Classification System); ASTM D471 (Effect of Liquids on Rubber); ASTM D395 (Compression Set of Rubber); ASTM D412 (Tensile Properties of Rubber); ASTM C1115; ASTM C361; ASTM C443; ASTM C564; ASTM C864; ASTM D1414; ASTM D1869; ASTM D5193; ASTM F104; ASTM F112; ASTM F1276; ASTM F146; ASTM F152; ASTM F2524; ASTM F36; ASTM F37; ASTM F38; ASTM F433; ASTM F495; ASTM	DIN ISO 3601-1 (O-Rings — Dimensions, Tolerances and Designation); DIN ISO 3601-5 (O-Ring Quality Acceptance Criteria); DIN ISO 815-1 (Compression Set of Rubber); DIN ISO 1817 (Resistance of Rubber to Liquids)	SAE J200 (Rubber Material Classification); API 6A (Oilfield Sealing Applications, where applicable); NORSOK M-710 (Oil & Gas Elastomer Qualification)

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
			F607; ASTM F806		
Pneumatic Seal	IS 9975 (Rubber Sealing Products); IS 14533 (Hydraulic & Pneumatic Seal Applications); IS 5382; IS 7466; IS 11149; IS 10188; IS 13137; IS 3400	ISO 5597 (Pneumatic Fluid Power — Rod Seal Housing Dimensions),ISO 3601-1; ISO 3601-3; ISO 3601-5; ISO 4633; ISO 6447; ISO 9631	ASTM D2000 (Rubber Products Classification System); ASTM D395 (Compression Set of Rubber); ASTM D412 (Tensile Properties of Rubber); ASTM D2240 (Durometer Hardness of Rubber); ASTM C1115; ASTM C361; ASTM C443; ASTM C564; ASTM C864; ASTM D1414; ASTM D1869; ASTM D5193; ASTM F104; ASTM F112; ASTM F1276; ASTM F146; ASTM F152; ASTM F2524; ASTM F36; ASTM F37; ASTM F38; ASTM F433; ASTM F495; ASTM F607; ASTM F806	DIN ISO 3601-1 (O-Rings for Fluid Power Applications); DIN ISO 815-1 (Compression Set); DIN ISO 37 (Tensile Stress-Strain Properties of Rubber); DIN ISO 188 (Accelerated Ageing Resistance)	SAE J200; Fluid Power Industry Specifications; OEM Pneumatic Equipment Requirements
Rotary Shaft Seal	IS 5129 (Rotary Shaft Lip Seals); IS 5382; IS 7466; IS 11149; IS 10188; IS 13137; IS 3400	ISO 6194-1 (Rotary Shaft Lip-type Seals Incorporating Elastomeric Sealing Elements — Nominal Dimensions and Tolerances); ISO 6194-5 (Identification of Visual Imperfections),ISO 3601-1; ISO 3601-3; ISO 3601-5; ISO 4633; ISO 6447; ISO 9631	ASTM D2000 (Rubber Products Classification System); ASTM D1414 (Rubber O-Ring and Seal Testing, where applicable); ASTM D471 (Effect of Liquids on Rubber); ASTM D395 (Compression Set of Rubber); ASTM C1115; ASTM C361; ASTM C443; ASTM C564; ASTM C864; ASTM D1869; ASTM D5193; ASTM F104; ASTM F112; ASTM F1276; ASTM F146; ASTM F152; ASTM F2524; ASTM F36; ASTM F37; ASTM F38; ASTM F433; ASTM F495; ASTM F607; ASTM F806	DIN 3760 (Rotary Shaft Lip Seals — Dimensions and Tolerances); DIN ISO 6194-1 (Rotary Shaft Lip-Type Seals — Nominal Dimensions and Tolerances); DIN ISO 815-1 (Compression Set); DIN ISO 1817 (Resistance to Fluids)	SAE J946 (Oil Seal Applications); RMA Rubber Handbook Classification; NSF/ANSI 61 (Water Applications, where applicable)
Oil Seal	IS 5129 (Rotary Shaft Lip Seals); IS 5382; IS 7466; IS 11149; IS 10188; IS 13137; IS 3400	ISO 6194-1 (Rotary Shaft Lip-type Seals — Nominal Dimensions and Tolerances); ISO 6194-4 (Performance Test Procedures),ISO 3601-1; ISO 3601-3; ISO 3601-5; ISO 4633; ISO 6447; ISO 9631	ASTM D2000 (Rubber Products Classification System); ASTM D1414 (Elastomeric Seal Test Methods); ASTM D471 (Effect of Liquids on Rubber); ASTM D412 (Tensile Properties of Rubber); ASTM C1115; ASTM C361; ASTM C443; ASTM C564; ASTM C864; ASTM D1869; ASTM D5193; ASTM F104; ASTM F112; ASTM F1276; ASTM F146; ASTM F152; ASTM F2524; ASTM F36; ASTM F37; ASTM F38; ASTM F433; ASTM F495; ASTM F607; ASTM F806	DIN 3760 (Rotary Shaft Seals); DIN ISO 6194-1 (Rotary Shaft Lip-Type Seals); DIN ISO 1817 (Resistance to Liquids); DIN ISO 188 (Ageing Resistance)	SAE J946 (Automotive Oil Seal Applications); SAE J200 (Rubber Material Classification); RMA Seal Standards
Mechanical Seal Elastomer	IS 9975 (Rubber Sealing Products); IS 14533 (Mechanical Seal Applications); IS 5382; IS 7466; IS 11149; IS 10188; IS 13137; IS 3400	ISO 21049 (Pumps — Shaft Sealing Systems for Centrifugal and Rotary Pumps); ISO 1817 (Rubber — Determination of Effect of Liquids),ISO 3601-1; ISO 3601-3; ISO 3601-5; ISO 4633; ISO 6447; ISO 9631	ASTM D2000 (Rubber Products Classification System); ASTM D1418 (Rubber Nomenclature); ASTM D471 (Effect of Liquids on Rubber); ASTM D395 (Compression Set of Rubber); ASTM C1115; ASTM C361; ASTM C443; ASTM C564; ASTM C864; ASTM D1414; ASTM D1869; ASTM D5193; ASTM F104; ASTM F112; ASTM F1276; ASTM F146; ASTM F152; ASTM F2524; ASTM F36; ASTM F37; ASTM F38; ASTM F433; ASTM F495; ASTM F607; ASTM F806	DIN EN 12756 (Mechanical Seals — Main Dimensions); DIN ISO 37 (Tensile Properties); DIN ISO 1817 (Resistance to Fluids); DIN ISO 815-1 (Compression Set)	API 682 (Mechanical Seals for Pumps in Petroleum, Petrochemical and Natural Gas Industries); NORSOK M-710 (Oil & Gas Elastomer Qualification); FDA 21 CFR 177 (Food Applications, where applicable)

Source field: Rubber and Rubber products